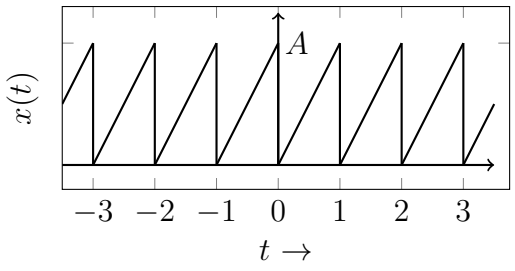
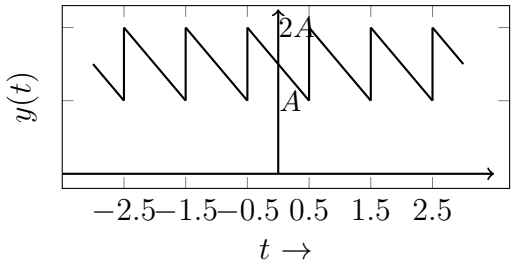
EE1101 Signals and Systems
Tutorial 5

1. A periodic signals $x(t)$ is given below.



   (a) Determine the Continuous Time Fourier Series (CTFS) coefficients of $x(t)$ in the exponential form. Sketch the magnitude and phase spectra.



   (b) Determine the CTFS coefficients of the periodic signal $y(t)$ in the exponential form. Sketch the magnitude and phase spectra.

   (c) Using Fourier series properties, derive the CTFS coefficients of $y(t)$ from those of $x(t)$.

2. Determine the CTFS coefficients of the following periodic signals with period $T = 1$. Sketch the magnitude and phase spectra.

   (a) $x_1(t) = \begin{cases} 1 & |t| < 1/4 \\ 0 & 1/4 \le |t| < 1/2 \end{cases}$

   (b) $x_2(t) = \begin{cases} \sin(2\pi t) & 0 \le t < 1/2 \\ 0 & -1/2 \le t < 0 \end{cases}$

3. A 2π - periodic signal $x(t)$ is specified over one period as

$$x(t) = \begin{cases} \dfrac{t}{A} & 0 \le t < A \\ 1 & A \le t < \pi \\ 0 & \pi \le t < 2\pi \end{cases}$$

   Represent $x(t)$ in its exponential Fourier series.

4. The exponential Fourier series coefficients of a periodic signal $x(t)$ are given by

$$a_k = \begin{cases} jk, & |k| < 3 \\ 0, & \text{otherwise} \end{cases}$$

   Furthermore, $x(t+4) = x(t)$. Determine the signal $x(t)$.

5. Let $x(t+T) = x(t)$, having exponential Fourier series coefficients a_k. Derive the Fourier series coefficients of the following signals in terms of a_k (Hint: Use analysis/synthesis equation).
   (a) $x(t - t_0)$
   (b) $x(-t)$
   (c) $x^*(t)$
   (d) $x(t - t_0) + x(t + t_0)$
   (e) Even$\{x(t)\}$
   (f) Real$\{x(t)\}$

6. Let $x(t)$ be a real-valued periodic signal with period T.
   (a) Show that if $x(t) = x(-t)$, the Fourier series expansion does not contain any sine terms.
   (b) Show that if $x(t) = -x(-t)$, the Fourier series expansion does not contain any cosine terms. What can you say about the value of a_0?

7. (a) Suppose the complex-valued periodic signal possesses conjugate symmetry, i.e, $x(t) = x^*(-t)$. What property do the FS coefficients possess?

   (b) Now suppose we define "evenness" for a *complex-valued* signal as x(t) = x(-t). Do the FS coefficients of x(t) possess any special property?

8. The trigonometric Fourier series expansion of real-valued periodic signal $x(t)$ with period T is given by,

$$x(t) = C_0 + \sum_{k=1}^{\infty} [C_k \cos(k\omega_0 t) + D_k \sin(k\omega_0 t)]$$

where,

$$C_0 = \frac{1}{T} \int_T x(t)\, dt$$
$$C_k = \frac{2}{T} \int_T x(t) \cos(k\omega_0 t)\, dt$$
$$D_k = \frac{2}{T} \int_T x(t) \sin(k\omega_0 t)\, dt$$

Find the trigonometric Fourier series coefficients of

(a) $x_1(t)$ in Q2(a)

(b) $x_2(t) = x_1(t - 0.25) - 0.5$

9. Find Fourier series coefficients of the periodic signal

$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$$

where $x(t) = x(t + T)$

10. Let $x(t) = |\sin t|$.

    (a) Plot the signals $x^{'}(t)$ and $x^{''}(t)$.

    (b) Find the Fourier series coefficients of the signal $x(t) + x^{''}(t)$.

    (c) Use the differentiation property to find the Fourier series coefficients of the signal $x(t)$.

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